

MULTI-LABEL TEXT CATEGORIZATION USING MACHINE LEARNING

INTERNSHIP PROJECT REPORT

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ABSTRACT

Multi-Label Text Categorization is a Natural Language Processing (NLP) and Machine Learning project designed to automatically classify text documents into multiple relevant categories. Unlike traditional classification systems that assign only one category, this system can assign multiple labels to a single text document. The project uses TF-IDF Vectorization and Logistic Regression to analyze textual content and predict appropriate categories such as Technology, Education, Health, and Sports.

INTRODUCTION

With the rapid growth of digital content, automatically organizing and categorizing text documents has become an essential task. Traditional text classification methods assign only one category to a document, whereas real-world documents often belong to multiple categories simultaneously.

This project implements a Multi-Label Text Categorization system using Machine Learning techniques. The model analyzes text data, converts it into numerical features, and predicts multiple categories for each document.

OBJECTIVES

- To develop a multi-label text classification system.
 - To apply Natural Language Processing techniques for text analysis.
 - To convert textual data into machine-readable features.
 - To predict multiple categories for a single document.
 - To improve automated content organization and management.
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TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Natural Language Processing (NLP)

SYSTEM REQUIREMENTS

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: 4 GB or higher
- Storage: 500 MB free space

Software Requirements

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas Library
- NumPy Library
- Matplotlib Library
- Scikit-learn Library

DATASET DESCRIPTION

The dataset consists of text documents along with their corresponding categories.

Sample Categories:

- Technology
- Education
- Sports
- Health

Each text document can belong to one or more categories, making it a multi-label classification problem.

METHODOLOGY

Step 1: Data Collection

A text dataset containing documents and corresponding category labels was created.

Step 2: Data Preprocessing

The text data was cleaned and prepared for machine learning analysis.

Step 3: Label Transformation

MultiLabelBinarizer was used to convert multiple category labels into binary format.

Step 4: Text Vectorization

TF-IDF (Term Frequency-Inverse Document Frequency) Vectorizer was used to convert text documents into numerical feature vectors.

Step 5: Data Splitting

The dataset was divided into training and testing sets.

Step 6: Model Training

A One-vs-Rest Logistic Regression classifier was trained using the TF-IDF features.

Step 7: Model Evaluation

The model performance was evaluated using:

- Accuracy Score
- Classification Report

Step 8: Text Prediction

The trained model predicts multiple categories for new text documents.

IMPLEMENTATION

The project was implemented using Python and Scikit-learn. Text documents were transformed into TF-IDF vectors and processed through a Logistic Regression classifier wrapped within a One-vs-Rest classification strategy. The system was trained using labeled text data and evaluated using standard machine learning metrics.

The implementation also includes keyword analysis, category distribution visualization, and prediction for new documents.

RESULTS

The system successfully:

- Classified text documents into multiple categories.
- Predicted categories for unseen text.
- Generated classification reports.
- Identified important keywords.
- Visualized category distribution.
- Performed multi-label classification effectively.

Example Predicted Categories:

- Technology
- Education
- Sports
- Health

ADVANTAGES

- Automatic document classification.
- Supports multiple categories per document.
- Improves content organization.
- Reduces manual classification effort.
- Scalable for large datasets.

APPLICATIONS

- News Article Classification
- Email Categorization
- Social Media Content Analysis
- Educational Content Management
- Document Management Systems
- Recommendation Systems

FUTURE ENHANCEMENTS

- Integration with large real-world datasets.
- Deep Learning-based text classification.
- Web application deployment.

- Real-time text categorization.
 - Support for multilingual text processing.
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CONCLUSION

The Multi-Label Text Categorization System successfully demonstrates the application of Machine Learning and Natural Language Processing in automated text classification. By using TF-IDF Vectorization and Logistic Regression, the system effectively predicts multiple categories for a given document. The project highlights the importance of NLP techniques in managing and organizing large volumes of textual information.

REFERENCES

1. Python Documentation
 2. Scikit-learn Documentation
 3. Pandas Documentation
 4. NumPy Documentation
 5. Matplotlib Documentation
 6. Natural Language Processing Research Papers
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